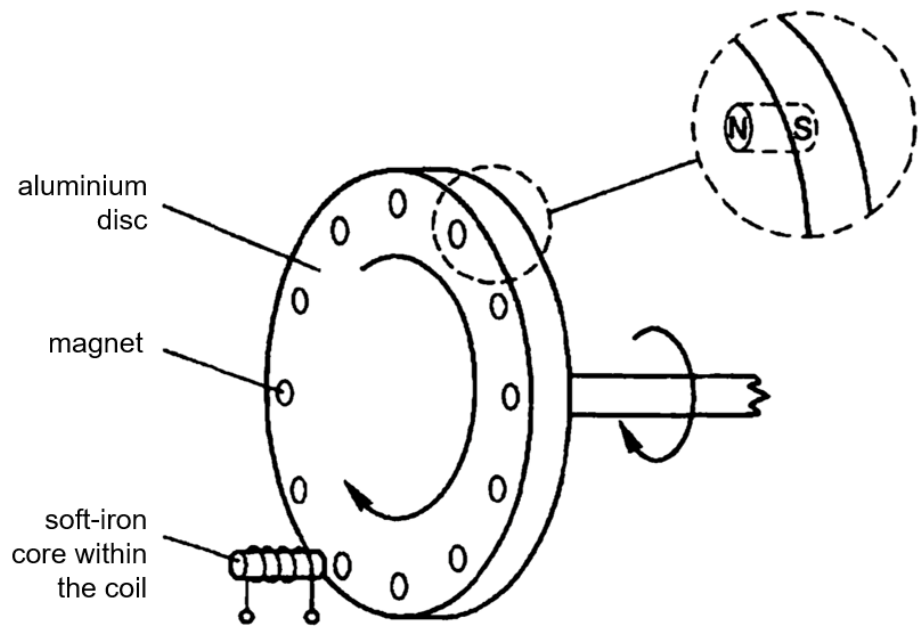
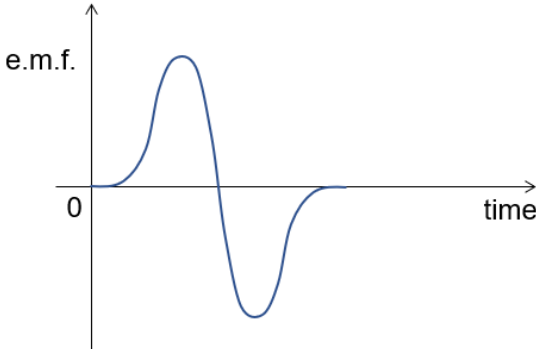


4	(a)	A revolving aluminium disc has small magnets equally spaced around its rim as shown in Fig. 4.1. The magnets are all aligned in the same direction with the north poles on the same side of the disc. The disc rotates at a constant angular velocity.  Fig. 4.1 A coil, wound on a soft-iron core, is fixed such that the north poles of the magnets pass close by the end of the coil without touching it. The terminals of the coil are connected to a detector which monitors the e.m.f. induced in the coil.
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		(i)	As one magnet passes the coil, use the laws of electromagnetic induction to explain	
			1.	why there is an induced e.m.f. in the coil.
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				
				

			2.	why there is a reversal in the direction of the induced e.m.f.	
					
					
					
					
					[1]
				By Lenz's law, the direction of the induced e.m.f in the time interval when the <u>magnetic flux linkage increases</u> will be negative to that in the time interval when the magnetic flux linkage <u>decreases</u>.	1
		(ii)		On Fig. 4.2, sketch a graph to show the variation with time of the e.m.f. induced in the coil as one magnet passes the coil. 	
				Fig. 4.2	[1]
				Solution: 	1
				1 mark for • General shape (accept other similar variation but not linear e.g. do not accept saw-tooth, do not accept plateau) • Start from e.m.f. = 0 (accept if graph did not start from $t = 0$)	

- (b) Fig. 4.3 and Fig. 4.4 show two views of a rectangular coil of height h and width d rotating with an angular speed ω about a vertical axis in a horizontal magnetic field of flux density B . At a certain instance of time t , the normal to the plane of the coil makes an angle of ωt with the magnetic field.

There are N turns in the coil.

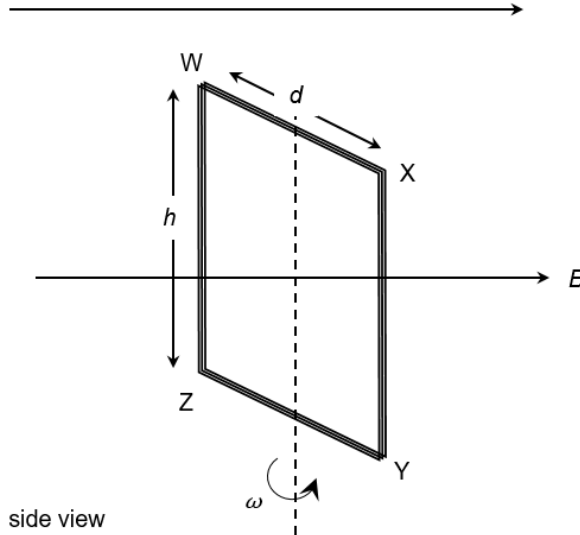


Fig. 4.3

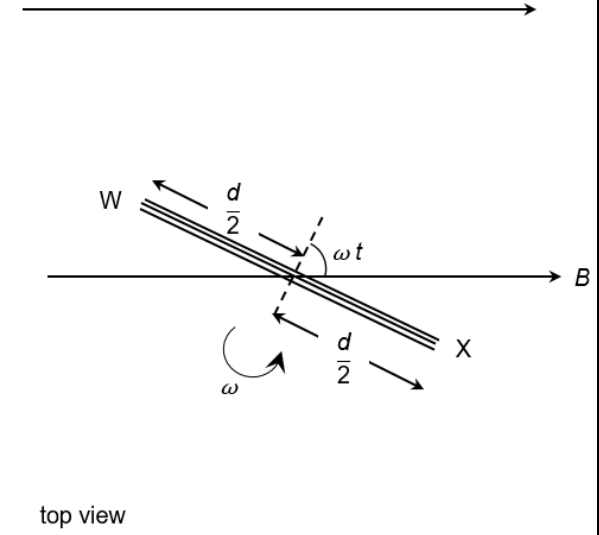


Fig. 4.4

- (i) At time $t = 0$, the plane of the coil is perpendicular to the magnetic field. Show that the magnitude of the induced e.m.f. E in the coil is given by

$$E = NBdh\omega \sin(\omega t)$$

[2]

Method 1

$$\begin{aligned} E &= -\frac{d\Phi}{dt} \\ &= -\frac{d(N\Phi)}{dt} \\ &= -N \frac{d(BA \cos \omega t)}{dt} \\ &= -NBA \frac{d(\cos \omega t)}{dt} \\ &= NBdh\omega \sin \omega t \end{aligned}$$

1

1

Method 2

The linear speed v of the sides XY and ZW as they execute circular motion about the vertical axis is given by

$$v = r\omega = \left(\frac{d}{2}\right)\omega$$

Hence, the e.m.f. induced in each of the sides XY and ZW is given by

$$E = NB_{\perp}lv = N(B \sin \omega t)lv = NBlv \sin \omega t$$

[1]

			$= NBh \left(\frac{d}{2} \right) \omega \sin \omega t$ Since the orientation of the induced e.m.f in XY and ZW are such that they are in the same direction, $E = E_{XY} + E_{ZW}$ $= 2 \left[NBh \left(\frac{d}{2} \right) \omega \sin \omega t \right]$ $= NBdh\omega \sin \omega t$		[1] [0]
		(ii)	The coil has dimension 30 cm by 24 cm and has 15 turns and the uniform magnetic field has flux density of 0.018 T. The coil rotates with a frequency of 25 Hz. Determine, for the coil,		
		1.	the maximum e.m.f. induced,		
			maximum e.m.f. induced = V		
			$\begin{aligned} \text{Max. e.m.f. induced, } E_0 &= NBdh\omega \\ &= NBdh(2\pi f) \\ &= 15(0.018)(24 \times 10^{-2})(30 \times 10^{-2})(2\pi(25)) \\ &= 3.053628 = 3.1 \text{ V} \end{aligned}$		
		2.	the root-mean-square value of the e.m.f. induced.		
			root-mean-square e.m.f. induced = V		
			$\begin{aligned} E_{RMS} &= \frac{E_0}{\sqrt{2}} \\ &= \frac{3.053628}{\sqrt{2}} \\ &= 2.1592 = 2.2 \text{ V} \end{aligned}$		

5 (a) Fig. 5.1 shows two small loudspeakers L_1 and L_2 separated by 15 cm. A microphone is